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METHODS PAPER

A simple method for measuring pollen germination rate using machine learning

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Abstract

The pollen germination rate decreases under various abiotic stresses, such as high-temperature stress, and it is one of the causes of inhibition of plant reproduction. Thus, measuring pollen germination rate is vital for understanding the reproductive ability of plants. However, measuring the pollen germination rate requires much labor when counting pollen. Therefore, we used the Yolov5 machine learning package in order to perform transfer learning and constructed a model that can detect germinated and non-germinated pollen separately. Pollen images of the chili pepper, *Capsicum annuum*, were used to create this model. Using images with a width of 640 pixels for training constructed a more accurate model than using images with a width of 320 pixels. This model could estimate the pollen germination rate of the F_2 population of *C. chinense* previously studied with high accuracy. In addition, significantly associated gene regions previously detected in genome-wide association studies in this F_2 population could again be detected using the pollen germination rate predicted by this model as a trait. Moreover, the model detected rose, tomato, radish, and strawberry pollen grains with similar accuracy to chili pepper. The pollen germination rate could be estimated even for plants other than chili pepper, probably because pollen images were similar among different plant species. We obtained a model that can identify genes related to pollen germination rate through genetic analyses in many plants.

Keywords Abiotic stress · Chili pepper · Microscope · Object detection · Yolov5

Introduction

Pollen fertility is essential for sexual plant reproduction and fruit production. Pollen fertility is reduced under various abiotic stress conditions, adversely affecting plant seed and fruit production. Therefore, measuring pollen fertility is crucial for agricultural production and plant research. Climate models predict that average temperature will continue to rise by 1.4 in Shared Socio-economic Pathway (SSP) scenario 1–1.9 or by 4.4 in SSP scenario 5–8.5 by the end of the twenty-first century (IPCC 2022). In many crops, plant reproductive ability, such as pollen fertility, is expected to decrease with global warming (Hedhly et al. 2009; Driedonks et al. 2016; Mesihovic et al. 2016), resulting in reduced yields. Thus, it has become increasingly essential to understand pollen fertility. In many cases, pollen fertility was evaluated by staining methods, such as fluorescein diacetate (FDA) and propidium iodide (PI), or acetocarmine staining. Measuring pollen germination rate was also the method for assessing pollen fertility, although the fertility by the staining method and pollen germination rate did not always match (Bolat and Pirlak 1999; Miller et al. 2021; Yamazaki et al. 2022).

The pollen germination rate can evaluate not only pollen fertility but also whether the pollen can be germinated under high temperatures. Therefore, in vitro pollen germination rates under high-temperature conditions have been investigated in many studies to assess the high-temperature tolerance of plants, such as *Gossypium hirsutum* (Kakani et al. 2005), *Arachis hypogaea* (Kakani et al. 2002), *Cocos nucifera* (Hebbar et al. 2018), *Oryza* spp. (Coast et al. 2016), *Glycine max* (Koti et al. 2005), and *Capsicum* spp. (Reddy and

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Kakani 2007; Gajanayake et al. 2011). Moreover, herbicide resistance has been assessed by in vitro pollen germination tests in *Lolium rigidum* (Richter and Powles 1993). Thus, an in vitro pollen germination test is an important tool to investigate stress tolerance, including high-temperature tolerance.

Evaluating pollen fertility and germination requires counting a lot of pollen, which is time-consuming and labor-intensive. Recently, a system for automatically investigating the various plant phenotypes of many plants using image analysis and machine learning has been constructed (Roitsch et al. 2019; Jiang and Li 2020). However, since a microscope is required for pollen observation, the automation of pollen fertility evaluation is challenging. With the development of microscope cameras and image analysis technology, there is an increasing number of cases in which trait surveys of plant microstructures are also conducted by image analysis. In the pollen fertility investigation with a staining method, a method of automating image analysis using ImageJ or a cell counter was developed (Ascari et al. 2020). On the other hand, only a few tools have been developed to automatically measure pollen germination rates (Güçlü et al. 2020; Yang et al. 2021), since it is more difficult to distinguish the presence or absence of pollen tubes than the difference in staining.

The most successful current approach to automated object detection is mainly based on deep learning. Object detection by deep learning is used in various fields including plant traits. For example, deep learning can detect diseases of different types of crops (Qi et al. 2022; Mathew and Yamuna 2022). Convolutional Neural Networks (CNNs) are deep learning algorithms that can extract high-level features of objects and can be effectively used for object detection and classification. The "you only look once" (Yolo) series is a one-stage object detection algorithm that uses CNNs as a backbone developed by Redmon et al. (2016). The Yolo series can predict multiple box positions and classes of each box. Several versions of the Yolo series have been developed (*v*1–*v*3) (Redmon et al. 2016; Redmon and Farhadi 2017; Redmon and Farhadi 2018). The Yolov5 model is an algorithm based on Yolov3 that improves detection speed while maintaining accuracy (Jocher et al. 2022; https://github.com/ultralytics/yolov5). Yolov5 consists of three main parts: the backbone, the neck, and the head. The backbone part was based on the New CSP-Darknet53. The neck part was based on the SPPF and the New CSP-PAN. The head part was designed the same as the Yolov3 head. Qi et al. (2022) reported that Yolov5 had a higher model detection speed, despite slightly lower accuracy for tomato disease detection compared with the target detection network of Faster R-CNN. Yolov5 was applied in this study for pollen germination due to its fast detection and training.

In the past, we conducted an association analysis of pollen germination rates under high temperatures in the F_2 population of *Capsicum chinense* (Yamazaki et al. 2020). At this time, it took a long time to investigate the pollen germination rate. Moreover, we must search for genetic resources and breed new cultivars with a high pollen germination rate under higher temperatures. Decreased pollen germination rate or pollen fertility due to abiotic stress, such as high-temperature, is a problem for many plants (Driedonks et al. 2016; Hedhly et al. 2009). Thus, it is necessary to investigate pollen germination rates in more plants. An automatic pollen germination measurement method was developed in *Capsicum* using machine learning and the Yolov5 model in this study. The trained model was applied to in vitro pollen germination tests in various crop species.

Materials and methods

Plant materials and the obtention of pollen germination images method for training models for detecting pollens

Chili pepper 'Takanotsume' and ornamental pepper 'Peruvian Purple' and their F_1 hybrid lines were used for the pollen germination test. We confirmed that these two cultivars have different pollen germination rates under high temperatures (Yamazaki et al. 2022). We used their images because we could obtain them from various pollen germination rates. The measurement of pollen germination rate on a soid medium followed the method of Yamazaki and Hosokawa (2019), with some modifications. Anthers from each flower were collected and transferred into a 1.5 mL plastic tube (five anthers from each individual). The pollens were suspended in 1 mL of a liquid solution composed of 5% (w/v) sucrose and 100 ppm (w/v) boric acid. A total of 0.2 mL of pollen suspension was dispersed on a 1% agar medium containing ingredients identical to those in the pollen suspension in the plastic Petri dishes (60 mm diameter) (Fig. 1A). The medium was placed in a dark incubator at approximately 25 °C. After 24 h, images of pollen germination were taken using a biological microscope (BX51; OLYMPUS, Tokyo, Japan) equipped with Visualix V900FL (Visualix, Kobe, Japan) or a DP71 camera (OLYMPUS) (Fig. 1B, C). The image size of pollen germination images was 2048 × 1080 pixels with Visualix V900FL or 1360 × 1024 pixels with a DP71 camera.

Pre-processing and annotation of image data for the training model

The image size was compressed to 320 × 169 or 320 × 241 pixels and 640 × 338 or 640 × 482 pixels to reduce the memory load of training. Each pollen in the image was tagged with "germinated" or "non-germinated" using Microsoft

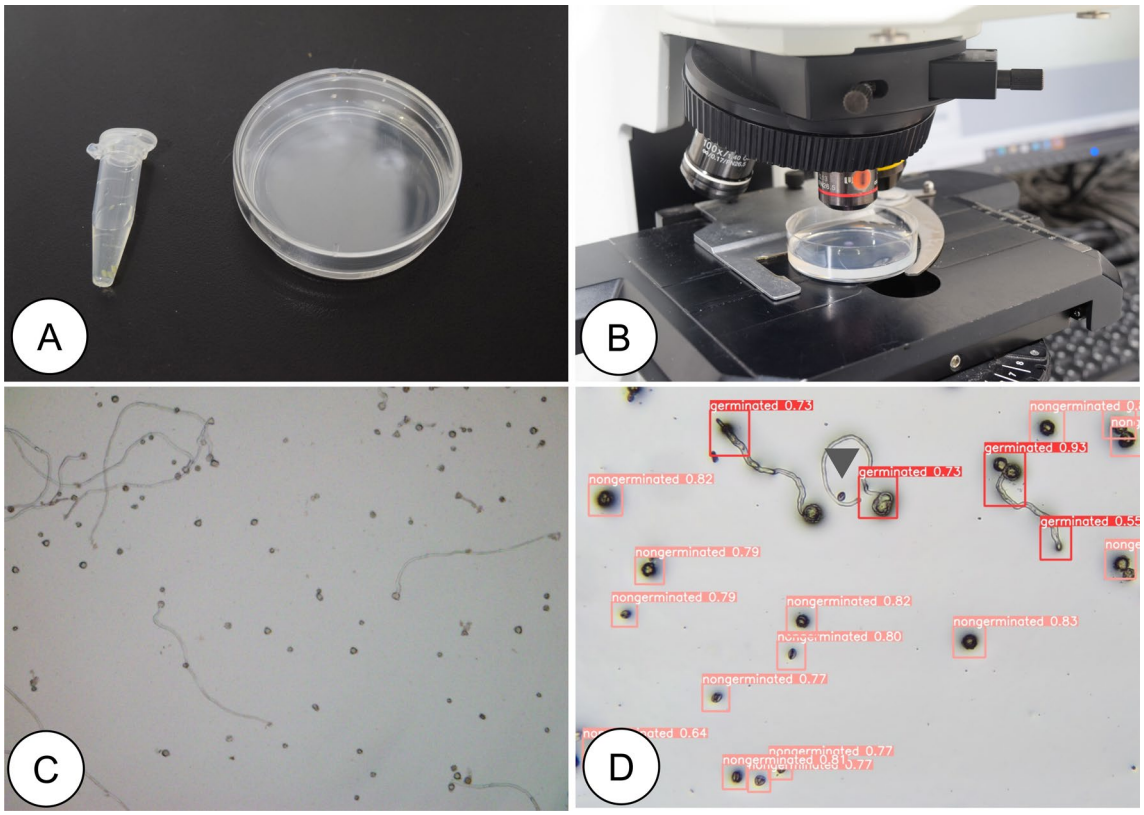


Fig. 1 Experimental method for obtaining pollen images. **A** Liquid medium and solid agar medium for pollen germination, **B** obtained pollen images on microscopy, **C** one of the pollen images, and **D** example of germinated and non-germinated pollen detected by Yolov5. Red squares represent predicted locations of germinated pollen. Pink squares represent predicted locations of non-germinated pollen. A triangle indicates aborted pollen not detected by the model as non-germinated pollen

VoTT (Microsoft, Redmond, USA) (Fig. S1). The annotation files were converted to a Yolo format using VoTT2Yolo (https://github.com/toritamantaro/vott2yolo_py). Two hundred images were used for annotation. The total number of annotations was about 1452 and 4297 for germinated and non-germinated pollen, respectively. Data augmentation was applied using yolo_augmentation (https://github.com/AR-Ray-code/yolo_augmentation). Data augmentation generated 24× images by creating image symmetry (2×), adding two types of noise (3×), and adjusting three types of brightness (4×). The total number of images in the training dataset was 4800 after data augmentation. Image and annotation data for the training model were divided into 3840 "training" images and 960 "validation" images. To avoid the risk of data leakage, we split the image data so that image data derived from the same image does not go into training and validation.

Training environment and evaluation indicators

The framework for deep learning was PyTorch 1.7.0. The experiment was run on Ubuntu 20.04.4 LTS, with an Intel Core i9-11900 K @ 3.50 GHz and an Nvidia GeForce RTX 3090 Super graphics card (NVIDIA, Santa Clara, USA). Transfer learning was performed by executing "train.py", which was included in the Yolov5 GitHub repository (https://github.com/ultralytics/yolov5), with the default settings. Yolov5 includes five subversions: yolov5n (nano), yolov5s (small), yolov5m (medium), yolov5l (large), and yolov5x (extra-large). The model of yolov5s was chosen to prevent the experimental computer from running out of memory. The batch size was 16, the resolution size of the training images was 320 pixels or 640 pixels, and the epochs were 500. Model training was stopped early if no improvement was observed in the last 100 epochs. Loss function and mean average precision (mAP) were calculated by default setting.

Evaluating a trained model

Two kinds of test datasets were prepared to evaluate the trained model. Dataset 1 concerned the examination of the precision of the pollen germination rate calculated from each image. Test dataset 1 consisted of 30 images of pollen germination, including 10 images of 'Takanotsume', 'Peruvian

Purple' and their F_1 hybrid individuals. The pollen grains of these cultivars were germinated, and images were obtained using the same method explained above. The pollen images were taken using a microscope BX51 equipped with a DP71 camera. The pollen germination rate was obtained by visually counting the number of germinated or non-germinated pollen grains. Pollen germination rates and numbers of pollen grains were used for the calculation of the root mean square error (RMSE). RMSE is one of the common metrics for validating model accuracy, and it helps in intuitive understanding of model accuracy because the smaller it is, the more accurate it is.

The other was test dataset 2, which concerned the evaluation of the precision of the pollen germination rate of an individual plant. Test dataset 2 included 10 individuals of the 'Takanotsume', 13 of the 'Peruvian Purple', 14 of the F_1 hybrid, and 65 of BC_1 population including F_1 hybrid × 'Takanotsume', F_1 hybrid × 'Peruvian Purple', and the 'Peruvian Purple' × F_1 hybrid. The pollen images were taken using a microscope BX51 equipped with a DP71 camera. Ten images were collected per individual with different microscopic fields on the same solid medium. The predicted pollen germination rate was calculated from the average of ten images of each individual. Pollen was detected from the pollen images executing "detect.py" included in the Yolov5 GitHub repository (Fig. 1D). The "best.pt" file output by executing "train.py" was used as a weight file for pollen detection. The "best.pt" file adopts the model obtained by the epoch with the highest weighted average of 1:9 of mAP@0.5 and mAP@0.5: 0.95 among epochs running train.py. Moreover, we used the save-txt function and output the result in text format. Actual numbers of germinated pollens were visually counted. Non-germinated pollen counts were measured with a script in Fiji (Schindelin et al. 2012), a software distribution of ImageJ (Schneider et al. 2012) using the "analyze particles" function. Actual pollen germination rates were calculated and used to calculate RMSE.

Application of the model for pollen of other crops

Tomato (*Solanum lycopersicum*), 'House-parto', strawberry (*Fragaria* × *ananassa*), 'Kaorino', rose (*Rosa* × *hybrida*), 'Fancy evening', Japanese radish (*Raphanus sativus*) and petunia (*Petunia* × *hybrida*), 'Surfinia yellow' were used for the pollen germination test. Pollen germination media were prepared according to the following reports: Masuda et al. (2010) reported that the pollen germination medium for tomatoes contained sucrose (100 g/L), H_3BO_3 (50 mg/L), and $CaCl_2$ $2H_2O$ (132.5 mg/L); Pipattanawong et al. (2009) reported that the medium for strawberry contained glucose (80 g/L) and H_3BO_3 (50 mg/L); Gudin et al. (1991) reported that the medium for rose contained H_3BO_3 (40 mg/L), $CaCl_2$ $6H_20$ (226.5 mg/L), and sucrose (150 g/L); Roberts et al. (1983) reported that the medium for Japanese radish contained sucrose (200 g/L), KNO_3 (0.1 g/L), $CaCl_2$ $2H_2O$ (243 mg/L), and H_3BO_3 (0.1 g/L); and Kovaleva et al. (2005) reported that the medium for petunia contained sucrose (150 g/L) and H_3BO_3 (0.1 g/L). All kinds of liquid medium were made into a solid medium containing 0.8% agar. The pollen was suspended in a liquid medium. A 0.1 mL pollen suspension was applied onto the solid medium using the same method as above. Pollen was germinated in the dark for 6–24 h. Pollen images of tomato and rose were taken using a microscope BX51 equipped with Visualix V900FL. Pollen images of strawberry, petunia and Japanese radish were taken using a CH30 microscope (OLYMPUS) equipped with Visualix STD1 (Visualix).

Application of the model for GWAS analysis

We previously detected gene regions associated with pollen germination rate under high temperatures in the *C. chinense* F_2 population (Yamazaki et al. 2020). Using the pollen germination image dataset from the previous study, we then analyzed whether the association with pollen germination rate could be detected in the same gene region using the machine learning method. There were 10 pollen germination images for each F_2 individual, and the pollen germination rate of 174 F_2 individuals was calculated. Pollen images of *C. chinense* were taken using a microscope BX53 (Olympus) equipped with DP73 (Olympus). A scatter plot was drawn between the actual pollen germination rate (values used in previous studies) and the predicted pollen germination rate (values calculated by machine learning in this study). In addition, an association analysis was performed using the same method proposed by Yamazaki et al. (2020), using the predicted pollen germination rate.

Results and discussion

Training model for pollen detection

The training was executed when the image size was compressed to a width of 320 or 640 pixels. The mAP, a general measure of model success, improved to 0.73 at images width of 320 pixels (Fig. S2) and 0.84 at images width of 640 pixels (Figs. 2, S3). We did not train on images larger than 640 pixels due to memory size. It was suggested that the size of the image in the training dataset might be important for a machine learning model to detect small pollen.

When training was applied to images with widths of 640 pixels, the validation loss did not improve after epoch 70 (Fig. S3). Therefore, it was considered that overfitting had occurred. On the other hand, both map@0.5 and map@0.5:0.95 were highest at epoch 52, producing the

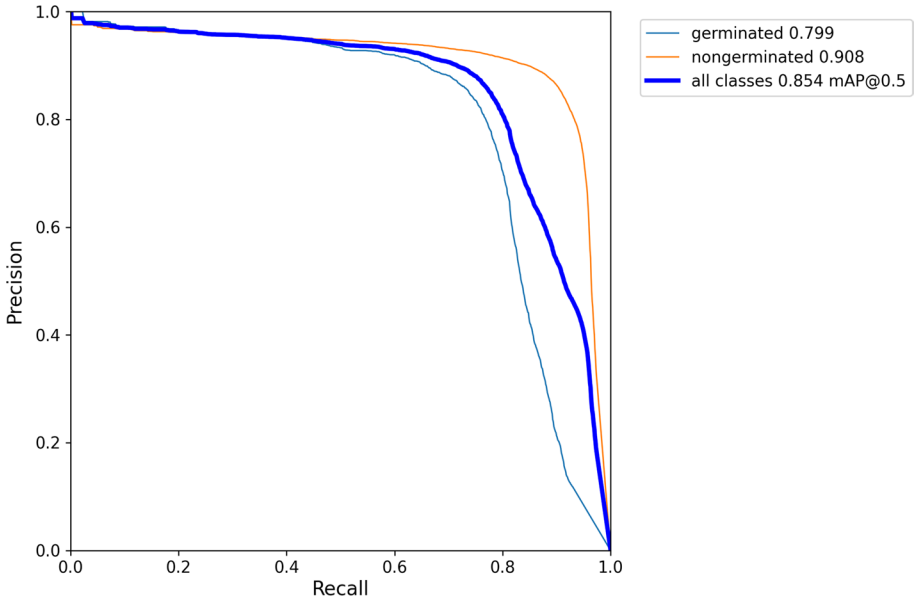


Fig. 2 Evaluation of the trained Yolov5 model. The PR curve represents precision on the vertical axis and recall on the horizontal axis. The figure was output by running "train.py" with default settings

weighted file "best.pt". This means that the model created at epoch 52 will be used for subsequent analysis. Thus, it would not be affected by overfitting. In addition, the number of epochs in which overfitting occurred for the training of the image with widths of 640 pixels tended to decrease compared with the image widths of 320 pixels.

Eighty-one percent of the true germinated pollen was detected correctly, while 95% of the true non-germinated pollen was detected correctly. We defined a false positive as the model falsely detecting the presence of pollen in a location where there is actually no pollen. Eighty percent of the false positives were detected as ungerminated pollen and 20% as germinated pollen (Fig. S4). Therefore, there was a difference in detectability between germinated and non-germinated pollen in this model. There was a tendency for non-germinated pollen to be particularly easy to detect. This is because the training data was unbalanced with more training data for non-germinated pollen than germinated pollen.

Test of the trained model in *Capsicum* pollen images

Whether germinated and non-germinated pollen could be detected by the model constructed using images with a width of 640 pixels was verified. Two test datasets that included image data not included in the "training" and "validation" data were used for verification. In test dataset 1, RMSE was 1.87, 9.53, and 4.21 at the germinated pollen number, non-germinated pollen number, and pollen germination rate, respectively (Fig. 3). In addition, the predicted and measured values were relatively close. In test dataset 2, the RMSE was 11.99, 250.23, and 13.91 in germinated pollen number, non-germinated pollen number, and pollen germination rate, respectively (Fig. S5A–C). Since the predicted number of non-germinated pollen was biased higher than that detected by "analyze particles" function of ImageJ, the value of the actual pollen germination rate was consisting lower than the predicted rate. Possibly, a small prediction is not because the model is flawed, but because there was pollen that could not be detected by the ImageJ script used to calculate the pollen germination rate. The actual pollen germination rate visually counted had a lower RMSE (4.84) with a predicted pollen germination rate than that counted with ImageJ (Fig. S5D–F). It was thought that it became possible to calculate the pollen germination rate with higher accuracy than recognizing the shape using the ImageJ script.

There was no large discrepancy between the previously measured value of pollen germination in Yamazaki et al. (2020) and the predicted germinated pollen number, non-germinated pollen number, and pollen germination rate by the model constructed in this study (Fig. 4A–C). The RMSE values between the actual and predicted germinated pollen number, non-germinated pollen number, and pollen germination rates were 28.8, 67.3, and 3.95, respectively. These plant materials are *C. chinense*, which is different from *C. annuum*, the species used for training data of pollen images. However, the dataset of *C. chinense* had a lower RMSE compared to the dataset 1 (*C. annuum*); the *C. chinense* dataset had an average of 9.8% for actual pollen germination, while the dataset 1 had an average of 16.1% (Fig. 3C), which was higher than.

C. chinense. This could be one of the reasons for the lower RMSE of *C. chinense*.

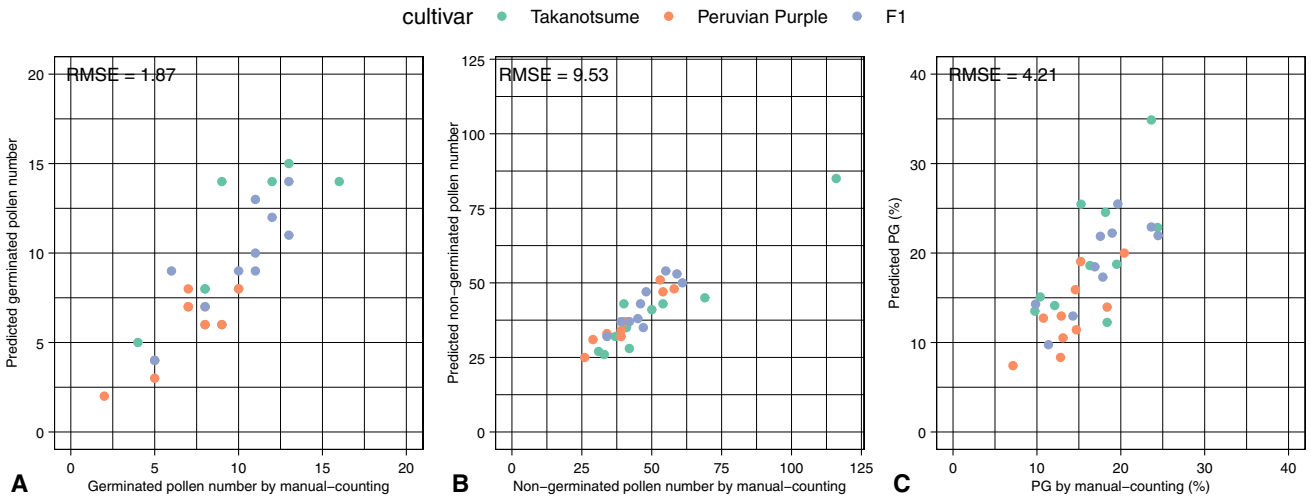


Fig. 3 Prediction of germinated pollen number (**A**), non-germinated pollen number (**B**), and pollen germination rate (**C**) for each image by the model constructed in this study. The vertical axis is the number of pollen and pollen germination rates detected from the images by the model constructed in this study. The horizontal axes of **A** and **B** are the number of germinated and non-germinated pollen counted by manual-counting. The pollen germination rate of **C** was calculated from these pollen numbers. Plots with different colors indicate different cultivars

The limitations of the model and possible sources of improvement

Some pollen may failed to fully develop (aborted) and not germinate (Fig. 1D). We did not separate non-germinated pollen into non-germinated pollen with developed normally and aborted pollen. It is likely that some aborted pollen is not recognized by the model, reducing the accuracy of the prediction of the number of non-germinated pollen. On the other hand, model accuracy is lower for germinated pollen than for non-germinated pollen (Figs. 3, 4, 5). Therefore, efficient detection of germinated pollen is also necessary to increase model accuracy. One possible reason for the low detection accuracy of germinated pollen is that the number of annotations for germinated pollen is low because images with a germination rate of less than 50% were used for training. Although we used constant imaging conditions in this study, the model could be more robust if images from various microscopes and imaging conditions are used in the transfer learning.

Application of pollen germination rates obtained by the model to GWAS analysis

To test whether it was possible to use the predicted pollen germination rate within the *Capsicum* genus for genetic analysis. Genome-wide association analysis was performed using the predicted pollen germination rate (Fig. 4C) as trait data. As a result, three gene regions detected a 5% significance level ($-\text{Log } P$ value > 4.88) (Fig. 4D). These gene regions were located in the same gene regions on chromosomes 2, 3, and 6 as in our previous study (Yamazaki et al. 2020). This result indicates that an association was detected in the same gene region as in previous studies. Therefore, it was shown that the model trained in this study could be used to identify genomic regions with potential influence on pollen germination. Some reports have detected gene regions associated with pollen germination rates (Ahmed et al. 2022), but the number is limited. One of the reasons is that a large amount of labor for investigating pollen germination rates was an obstacle to gene identification. Ascari et al. (2020) have enabled rapid pollen viability evaluation with the dye staining method in many plant species using image analysis software, ImageJ Fiji and CellProfiler. Furthermore, a method for evaluating pollen viability using Yolov5 was reported in *Citrus* (Komiya and Goto 2022). However, to date, there has not been a robust image analysis method for pollen germination. Pollen viability with the dye staining method facilitates image analysis software or script because it can be distinguished by color. In contrast, pollen germination is challenging to detect due to the difficulty of pollen tubes. It was suggested that the tools trained in this study, Yolov5, would be helpful for future genetic studies on pollen germination.

Application of the model to the pollen of other crops

The model was trained using pollen images of chili pepper, *C. annuum*. We were able to predict the pollen germination rate of *C. chinense* pollen germination images with the same accuracy as *C. annuum* images using this method

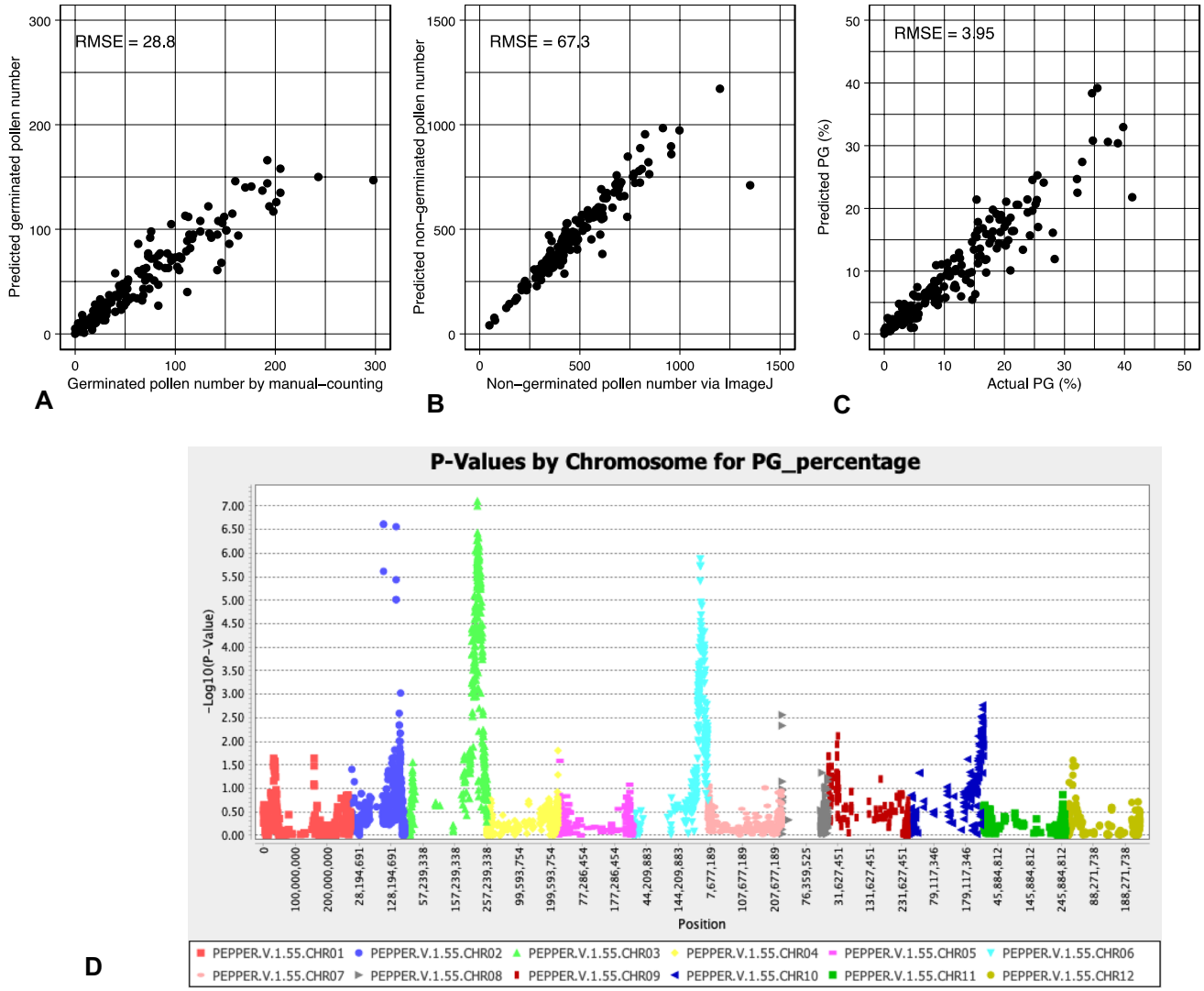


Fig. 4 Application of pollen germination rate measurement by Yolov5 to GWAS analysis. Comparison of germinated pollen number (**A**), non-germinated pollen number (**B**), and pollen germination rate (**C**) predicted by the model. The vertical axis is the number of pollen of **A** and **B** and pollen germination rates of (**C**) detected from the image by the model constructed in this study. The measured data were investigated by Yamazaki et al. (2020). The horizontal axis of **A** is the number of germinated pollens counted by manual-counting. The horizontal axis of **B** is the number of non-germinated pollens counted by the "analyze particles" of ImageJ. The number of germinated and non-germinated pollens was calculated from ten images per individual. The pollen germination rate of **C** was calculated from the pollen numbers. **D** GWAS analysis with pollen germination rate predicted by the model. Genotype data were investigated by Yamazaki et al. (2020). Different color plots show SNPs on 12 chromosomes. The vertical axis shows the *P* value (−Log10) associated with the pollen germination rate. The *P* value was corrected by Bonferroni's method, and correlations of $P<0.01$ and $P<0.05$ were obtained when the vertical axis was 5.58 or more and 4.88 or more, respectively

(Figs. S5C, 4C). Therefore, we hypothesized that it could be used in other plant species and verified whether the model trained in this study could be used in tomatoes, strawberries, radishes, and roses (Fig. S6). The RMSE values were 7.82, 6.65, 2.89 and 5.83 in tomato, strawberry, rose and petunia, respectively (Fig. 5, Table 1). On the other hand, a dataset that could be used to predict the pollen germination rate could not be obtained in radish since germinated pollen was hardly obtained (RMSE = 10.93). However, the RMSE of the predicted non-germinated pollen with the actual value was 4.32, and the pollen could be detected with relatively high accuracy. A lower RMSE was obtained for roses of the Rosaceae family than for tomatoes, which belong to the same Solanaceae family as *C. annuum*. Therefore, it was not difficult to predict the pollen germination rate, even without a close genetic relationship to *C. annuum*. This model has the potential to predict pollen germination rates in a wide range of plants. In addition, as a result of using different microscopes and cameras to acquire the pollen image, the size of the image and the shooting conditions were different.

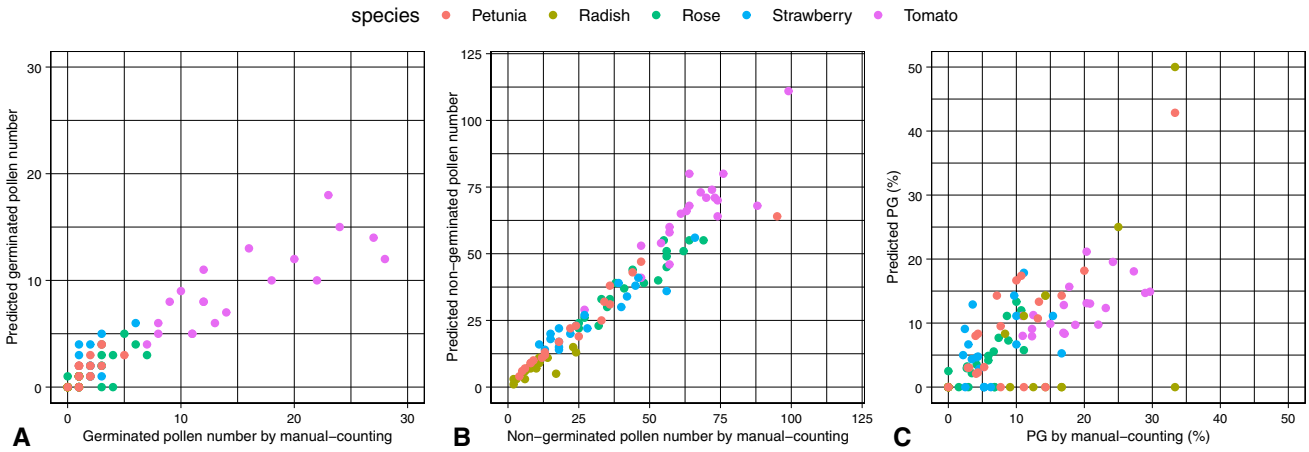


Fig. 5 Comparison between the predicted and actual germinated pollen number (**A**), non-germinated pollen number (**B**), and pollen germination rate (**C**). The vertical axis is the number of pollen of (**A**) and (**B**) and pollen germination rates of (**C**) detected from the image by the model constructed in this study. The horizontal axis of (**A**) and (**B**) is the number of germinated and non-germinated pollens counted by manual-counting. The pollen germination rate of (**C**) was calculated from the pollen numbers. Petunia, Japanese radish, rose, strawberry, and tomato were plotted in orange, khaki, green, blue, and pink, respectively. Pollen germination rates were calculated for each image. $n = 20$

Table 1 RMSE values between predicted and actual germinated pollen numbers, non-germinated pollen numbers, and pollen germination rates for each species

Species	Germinated pollen number	Non-germinated pollen number	Pollen germination rate
Rose	1.64	7.02	2.89
Tomato	7.21	7.79	7.82
Japanese radish	0.55	4.28	10.93
Strawberry	1.28	6.64	6.65
Petunia	0.95	7.43	5.83

However, pollen was detected, and the pollen germination rate could be estimated. Therefore, it is considered that the model trained in this study may apply to a wide range of plants and laboratory equipment. Pollen images taken with microscope cameras from Olympus and Visualix verified the accuracy of the measurements in this study. Further research is needed to determine if images from different type of camera including cell phone cameras, digital cameras, or microscope cameras from other manufacturers can be used to measure pollen germination rates by this model.

Author's contribution statements AY and AT performed experiments and data analysis. AY, AT and KN (Kyoka Nagasaka) wrote the manuscript. KM and KN (Kazasa Nishimura) supported data analysis and writing manuscript. RN and TN advised on the experimental design and manuscript. All authors approved the final version of the manuscript..

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00497-023-00472-9.

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Data availability The code file in which Yolov5 was executed in this study and the R code for summarizing the data are being made available as supplementary material. The weight file "best.pt" and the training dataset including pollen images and annotations used in this study are available on our GitHub repository (https://github.com/AAkiraYamazaki/PollenCounter) and available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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